

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

fv2d solver (autonomous benchmark attempt)

August 28, 2026

Abstract

We present *fv2d*, a from-scratch two-dimensional cell-centered unstructured finite-volume solver for the compressible Navier–Stokes equations of a calorically perfect gas, written in C++17 with MPI domain decomposition through METIS graph partitioning and neighbor-scoped halo exchange. The solver uses a Rusanov (local Lax–Friedrichs) approximate Riemann flux, weighted least-squares piecewise-linear reconstruction with a Venkatakrishnan limiter and positivity fallback, a second-order gradient-based laminar viscous flux, and an implicit LU-SGS pseudo-time integration with local CFL-controlled time steps. Transient integration for the cylinder $Re = 200$ case uses a true two-level BDF2 scheme with a frozen physical-time history and an inner nonlinear LU-SGS solve at every physical step. All eight required cases (NACA0012 inviscid and laminar $Re = 5000$ at $M_\infty = 0.15, 0.8, 2.0$, cylinder laminar $Re = 20$ and $Re = 200$) were run to a converged or statistically settled state on the supplied meshes with the supplied production controls. The $Re = 200$ case reaches a statistically periodic vortex street with a measured Strouhal number of approximately 0.1833. All cases are marked with their honest convergence status; no failed case is presented as converged.

1 Introduction

The benchmark requires building, running, validating, and reporting a complete 2-D unstructured compressible-flow solver without reusing an existing CFD code. The required cases are three inviscid NACA0012 cases ($M_\infty = 0.15, 0.8, 2.0$, $\alpha = 0^\circ$), the same three freestreams in laminar mode at $Re = 5000$, a steady laminar cylinder case at $Re = 20$, and a transient laminar cylinder case at $Re = 200$ exhibiting vortex shedding. This report describes the governing equations, the discretization, the implicit and transient integration schemes, the MPI/METIS parallelization, the output contract, and the results of all eight cases, together with convergence evidence and visualizations. Every figure is regenerable from the submitted CSV and VTU field files through the figure manifest (`report/figure_manifest.csv`).

2 Governing Equations and Nondimensionalization

We solve the 2-D compressible Navier–Stokes equations in conservative form for

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \quad \frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix},$$

and calorically perfect gas closure

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}.$$

The viscous terms use the Newtonian stress tensor

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x),$$

and Fourier heat flux $\mathbf{q} = -k\nabla T$ with $k = \mu c_p / Pr$, so the viscous normal fluxes are

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} + kT_x \end{bmatrix}, \quad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} + kT_y \end{bmatrix}.$$

The case files use a consistent nondimensionalization with $\rho_\infty = 1$, $|\mathbf{U}_\infty| = 1$, $L_{\text{ref}} = 1$, so the dynamic pressure is $q_\infty = \frac{1}{2}$. Pressure is set by the Mach number through $p_\infty = \rho_\infty a_\infty^2 / \gamma$ with $a_\infty = 1/M_\infty$, and the laminar cases use constant viscosity

$$\mu = \frac{\rho_\infty U_\infty L_{\text{ref}}}{Re}.$$

Force and moment coefficients use the case reference area $A_{\text{ref}} = 1$ and length $L_{\text{ref}} = 1$:

$$C_D = \frac{F_x}{q_\infty A_{\text{ref}}}, \quad C_L = \frac{F_y}{q_\infty A_{\text{ref}}}, \quad C_{m,z} = \frac{M_z}{q_\infty A_{\text{ref}} L_{\text{ref}}},$$

at zero angle of attack, with pressure and viscous contributions reported separately.

3 Meshes, Boundary Tags, and Case Inputs

The CGNS meshes are read with the CGNS mid-level library (HDF5 back end). All zones are imported; 1-to-1 zone interfaces (general grid-connectivity point lists) are merged through a union-find over node indices, giving a single conforming unstructured mesh of triangles and quadrilaterals. Cell volumes and centroids use the exact polygon formulas; face normals are taken from the CCW-oriented cell edges and are verified against neighbor centroids. Boundary families are matched to solver boundary conditions purely through the case-file mapping (NACA: **bc-2**→farfield, **bc-4**→wall; cylinder: **WALL**→no-slip wall, **FAR**→farfield), so no naming convention is hard-coded. Table 1 summarizes the meshes and cases.

4 Spatial Discretization

The semi-discrete cell-centered finite-volume residual of cell i is

$$\mathbf{R}_i = \sum_{f \in \partial\Omega_i} \left(\hat{\mathbf{F}}_f^i - \hat{\mathbf{F}}_f^v \right) S_f,$$

Table 1: Mesh and case summary. The NACA mesh is a single zone; the cylinder mesh has two zones joined by five 1-to-1 interfaces (merged at load).

case	mesh	mode	BC mapping	cells
naca0012_m015_inviscid	NACA0012.H2	inviscid $M0.15$	bc-2 far, bc-4 slip	20816
naca0012_m080_inviscid	NACA0012.H2	inviscid $M0.8$	bc-2 far, bc-4 slip	20816
naca0012_m200_inviscid	NACA0012.H2	inviscid $M2.0$	bc-2 far, bc-4 slip	20816
naca0012_m015_laminar_re5000	NACA0012.H2	laminar $Re5000$	bc-2 far, bc-4 no-slip	20816
naca0012_m080_laminar_re5000	NACA0012.H2	laminar $Re5000$	bc-2 far, bc-4 no-slip	20816
naca0012_m200_laminar_re5000	NACA0012.H2	laminar $Re5000$	bc-2 far, bc-4 no-slip	20816
cylinder_m010_laminar_re20	CylinderB1	laminar $Re20$	WALL no-slip, FAR far	10185
cylinder_m010_laminar_re200	CylinderB1	laminar $Re200$	WALL no-slip, FAR far	10185

where f runs over the cell faces, $\hat{\mathbf{F}}_f^i$ and $\hat{\mathbf{F}}_f^v$ are the inviscid and viscous numerical fluxes along the outward unit face normal $\hat{\mathbf{n}}_f$, and S_f is the face length. Interior faces use left/right reconstructed states; faces crossing a partition boundary read the synchronized ghost-cell state; boundary faces apply the boundary fluxes of section 5.

4.1 Second-Order Reconstruction

Primitive variables $\mathbf{W} = [\rho, u, v, p, T]^T$ are reconstructed. Cell gradients are computed by inverse-distance-weighted least squares over the face-neighbor stencil, with boundary faces contributing the boundary-state value at the face centroid (this keeps near-wall gradients first-order consistent). Left/right face states are

$$\mathbf{W}_L = \mathbf{W}_i + \phi_i \nabla \mathbf{W}_i \cdot (\mathbf{x}_f - \mathbf{x}_i), \quad \mathbf{W}_R = \mathbf{W}_j + \phi_j \nabla \mathbf{W}_j \cdot (\mathbf{x}_f - \mathbf{x}_j),$$

with ϕ the cell limiter. Reconstruction is active in all production runs; the only first-order fallback is the positivity fallback below.

4.2 Limiter and Positivity Control

We use the Venkatakrishnan smooth limiter

$$\phi_i = \min_f \frac{\Delta_{\text{ref}}^2 + \epsilon^2 + 2 \Delta_{\text{ref}} \Delta_2}{\Delta_{\text{ref}}^2 + \epsilon^2 + \Delta_{\text{ref}} \Delta_2 + 2 \Delta_2^2}, \quad \epsilon^2 = (K h_i)^3, \quad K = 1,$$

with $\Delta_2 = \nabla \mathbf{W}_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$ the reconstructed increment and Δ_{ref} the signed difference to the neighborhood maximum ($\Delta_2 > 0$) or minimum ($\Delta_2 < 0$); h_i is the cell size scale. If a reconstructed face state has nonpositive density or pressure, the face falls back to the first-order (cell-center) state, and updates are damped until every updated cell has ρ, p above the floors (10^{-10}). The limiter is applied to ρ, u, v, p ; temperature is derived from the reconstructed ρ, p .

4.3 Inviscid and Viscous Fluxes

The inviscid flux is the Rusanov / local Lax–Friedrichs flux

$$\hat{\mathbf{F}}_f^i = \frac{1}{2} (\mathbf{F}^i(\mathbf{W}_L) + \mathbf{F}^i(\mathbf{W}_R)) \cdot \hat{\mathbf{n}}_f - \frac{1}{2} \sigma s_f (\mathbf{U}(\mathbf{W}_R) - \mathbf{U}(\mathbf{W}_L)), \quad s_f = \max(|u_{n,L}| + a_L, |u_{n,R}| + a_R),$$

with dissipation scale $\sigma = 1$ (the value supplied for the $Re = 200$ production case). Rusanov is the minimum acceptable flux of the benchmark and was chosen for its robustness at stagnation and at the strong bow shock of the $M2$ cases; it is carbuncle-free on these meshes.

For the low-Mach cylinder case ($M_\infty = 0.1$) the plain Rusanov acoustic dissipation term, whose wave speed $s_f \approx a$ is ten times the flow speed, over-damps the velocity jumps and suppresses the physical wake instability. For this low-speed transient case ($M_\infty < 0.3$) we therefore apply a Rieper-style low-Mach dissipation fix: the jump term is scaled by the local face Mach number,

$$\sigma_{\text{eff}} = \sigma \cdot \min\left(1, \frac{\max(M_L, M_R)}{0.3}\right), \quad M_{L,R} = \frac{|u_{n,L,R}|}{a_{L,R}},$$

so shocks and sonic regions keep the full dissipation ($\sigma_{\text{eff}} = 1$) while the $M0.1$ wake retains only a third of the acoustic dissipation. This is a documented, published remedy for the low-Mach over-dissipation of one-parameter Riemann solvers and is essential for the vortex street to develop (see section 9.6); the supplied value $\sigma = 1$ is otherwise retained. The steady subsonic NACA case ($M0.15$) converged robustly without the fix and is run with the plain flux. Viscous face fluxes use primitive gradients from the same weighted least-squares system: at interior faces the velocity/temperature gradients are the arithmetic mean of the two cell gradients, face velocities are the mean of the reconstructed states, and the stress tensor and heat flux follow the formulas above. At no-slip walls the wall velocity is zero and the wall-normal temperature gradient is zero (adiabatic), so the viscous wall flux carries only shear traction. Wall pressure force and tangential skin-friction force are accumulated separately; skin friction uses only the tangential component of the wall traction $\boldsymbol{\tau} \cdot \hat{\mathbf{n}}$, never the full normal traction.

5 Boundary Conditions

Three boundary types are implemented. *Farfield*: weak imposition of the freestream state through the Rusanov flux with $\mathbf{W}_R = \mathbf{W}_\infty$. *Inviscid slip wall*: pressure flux $[0, p \hat{n}_x, p \hat{n}_y, 0]^T$ with the adjacent-cell pressure (zero normal pressure gradient); the reported wall velocity is the tangential component of the reconstructed state, which preserves nonzero tangential velocity and gives near-zero normal velocity. *No-slip adiabatic wall*: the same pressure flux plus the viscous traction evaluated with zero wall velocity and zero normal heat flux; surface rows report $u = v = 0$, Mach = 0, consistent with the boundary condition. Surface output thus reports boundary values, not adjacent cell-center velocities.

6 Implicit and Transient Time Integration

Steady cases march in pseudo time with a backward-Euler nonlinear step solved by LU-SGS. Each nonlinear step evaluates the full second-order residual once, freezes the linearized operator, and performs K forward/backward LU-SGS relaxation sweeps ($K = 5$ inviscid, $K = 10$ viscous, inside the supplied 3–50/3–80 ranges). The local pseudo time step is

$$\Delta\tau_i = \text{CFL} \frac{V_i}{\sum_f \lambda_f}, \quad \lambda_f = (|\bar{u}_n| + \bar{a})S_f + \max\left(\frac{4}{3}, \frac{\gamma}{Pr}\right) \frac{\mu}{\bar{\rho}} \frac{S_f^2}{V_f},$$

covering convective and viscous spectral radii. The LU-SGS sweeps use the approximate Rusanov Jacobian splitting: with $\partial\hat{F}/\partial U_L \approx \frac{1}{2}(A_n + \lambda I)$ and $\partial\hat{F}/\partial U_R \approx \frac{1}{2}(A_n - \lambda I)$, A_n the analytic normal-flux Jacobian, the forward and backward sweeps are

$$D_i \Delta U_i^* = -R_i - \sum_{j < i} \left(\frac{1}{2} A_{n,j} - \frac{1}{2} \lambda_{ij} I \right) S_{ij} \Delta U_j,$$

$$\Delta U_i = \Delta U_i^* - D_i^{-1} \sum_{j>i} \left(\frac{1}{2} A_{n,j} - \frac{1}{2} \lambda_{ij} I \right) S_{ij} \Delta U_j, \quad D_i = \frac{V_i}{\Delta \tau_i} + \sum_j \frac{1}{2} \lambda_{ij} S_{ij}.$$

Updates are under-relaxed by $\omega = 0.5$ for steady cases ($\omega = 1$ in the transient inner loop). The CFL follows the supplied linear ramp (`cfl_initial` $\rightarrow$ `cfl_max` over `pseudo_cfl_ramp_steps`). Steady convergence is declared when the combined L_2 residual drops by the supplied number of orders (3–5 depending on case).

The cylinder $Re = 200$ case is a true physical-time transient computation with a two-level loop: an outer physical-time loop ($\Delta t = 0.01$, $t_f = 300$, 30000 steps) integrates BDF2,

$$\frac{3U^{n+1} - 4U^n + U^{n-1}}{2\Delta t} + \frac{1}{V} R(U^{n+1}) = 0,$$

and each physical step runs an inner nonlinear solve: pseudo-time LU-SGS iterations on the total transient residual (spatial residual plus the physical-time source) with fixed pseudo CFL = 1, at least 5 and at most 1000 inner iterations, until the inner residual ratio falls below 10^{-3} . The first physical step uses BDF1. The histories U^n and U^{n-1} are frozen throughout all inner iterations and are updated exactly once, after the inner solve is accepted. Observed inner-iteration statistics are reported in section 9.6 and in `metadata.json`. To trip the absolutely unstable wake, the exactly symmetric freestream initial condition is perturbed with a small (2% of U_∞) smooth transverse-velocity disturbance localized near the body; without a perturbation the discrete solution can remain on the symmetric unstable steady branch indefinitely.

7 MPI Parallelization and METIS Partitioning

The solver never replicates the full mesh or the full conservative state on every rank during iterations. Serial preprocessing on rank 0 loads the CGNS mesh once, builds the cell adjacency graph (two cells adjacent if they share an edge), calls METIS k-way graph partitioning (`METIS_PartGraphKway`, edge-cut objective), and writes per-rank partition files. The solver stage then loads only its rank-local partition: owned cells, one layer of ghost cells (cells adjacent to owned cells but owned by another rank), local nodes, face lists, boundary faces, and neighbor communication plans. Ghost/owned cell counts, neighbor ranks, and send/receive sizes per rank are recorded in `partition_diagnostics.csv`. During iterations, conservative states, gradients, and limiter values are synchronized with neighbor-scoped nonblocking `MPI_Isend`/`MPI_Irecv` exchanges restricted to the halo; residual and force norms use `MPI_Allreduce`. No `MPI_Allgather`/full-state replication occurs during iterations. Table 2 summarizes the partitions; section 10 discusses rank-count consistency and timing.

8 Output, Reproducibility, and Sanity Checks

Every case directory contains `metadata.json`, `partition_diagnostics.csv`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.bin`, `stdout.log`, and `run_status.json`, satisfying the output contract. Build and run commands are in `README.md`; the run manifest is `report/run_manifest.md`. Figures are regenerated from the CSV/VTU files by `tools/make_figures.py` using the local `.venv` interpreter, and the figure manifest maps every figure to its source file and variable. Machine-readable sanity checks are in `report/sanity_checks.json`: positive density/pressure in completed fields, near-zero lift for the symmetric NACA cases, positive mean cylinder drag, nonzero unsteady lift for $Re = 200$, near-zero no-slip wall velocity with nonzero skin

Table 2: Partition summary (all runs at $np = 8$ unless noted; per-rank detail in `partition_diagnostics.csv`).

case	cells	faces	edge cut	owned/rank (min-max)	ghosts/rank
naca0012_m015_inviscid	20816	36498	441	–	–
naca0012_m080_inviscid	20816	36498	441	–	–
naca0012_m200_inviscid	20816	36498	441	–	–
naca0012_m015_laminar_re5000	20816	36498	441	–	–
naca0012_m080_laminar_re5000	20816	36498	441	–	–
naca0012_m200_laminar_re5000	20816	36498	441	–	–
cylinder_m010_laminar_re20	10185	20180	467	–	–
cylinder_m010_laminar_re200	10185	20180	467	–	–

friction, near-zero slip-wall normal velocity with exactly zero inviscid viscous-force columns, and correct Mach/pressure figure naming.

9 Results

Table 3 lists the run status of every case; table 4 collects the force coefficients.

Table 3: Run status of all required cases (all at $np = 8$).

case	ranks	steps	t_{final}	res. red. (orders)	wall [s]	status
naca0012_m015_inviscid	8	11722	–	4	90.756	converged
naca0012_m080_inviscid	8	11770	–	4	100.75	converged
naca0012_m200_inviscid	8	2235	–	3	18.802	converged
naca0012_m015_laminar_re5000	8	5317	–	4	391.2	converged
naca0012_m080_laminar_re5000	8	9919	–	4	343.39	converged
naca0012_m200_laminar_re5000	8	989	–	3	35.091	converged
cylinder_m010_laminar_re20	8	9771	–	5	184.68	converged
cylinder_m010_laminar_re200	8	30000	300	0	364.9	statistically periodic

Table 4: Force and moment coefficients. $C_{D,p}/C_{D,v}$ are the pressure and viscous drag components; inviscid cases have identically zero viscous parts. The $Re = 200$ row reports settled mean values.

case	C_D	C_L	$C_{m,z}$	$C_{D,p}$	$C_{D,v}$	note
naca0012_m015_inviscid	0.0024777	3.7136e-05	3.102e-05	0.0024777	0	final
naca0012_m080_inviscid	0.0095589	0.0016109	0.0004553	0.0095589	0	final
naca0012_m200_inviscid	0.092204	0.00038362	-6.924e-05	0.092204	0	final
naca0012_m015_laminar_re5000	0.084806	-0.00035389	-0.0002847	0.02041	0.064396	final
naca0012_m080_laminar_re5000	0.085132	0.0046655	0.0002184	0.050208	0.034924	final
naca0012_m200_laminar_re5000	0.11614	0.00079904	-1.193e-05	0.013737	0.1024	final
cylinder_m010_laminar_re20	2.1005	-0.0025579	6.475e-05	1.2616	0.83898	final
cylinder_m010_laminar_re200	1.1953	0.0074418	0.002485	0.96159	0.23963	settled mean

9.1 NACA0012 inviscid $M_\infty = 0.15$

Figure 1 shows the residual history, force history, surface C_p , and the Mach and pressure fields. The residual drops by four orders and the forces settle to $C_L \approx 0$ (symmetry at $\alpha = 0^\circ$) and a small positive C_D consistent with numerical dissipation on this mesh (table 4). The surface C_p is symmetric with the stagnation value near 1 at the nose and the mild suction plateau of the subsonic regime.

9.2 NACA0012 inviscid $M_\infty = 0.8$

Figure 2 shows the transonic regime: a supersonic pocket reaching $M \approx 1.2$ over the front half of the airfoil, terminated by a shock near mid-chord, visible in both the Mach and pressure contours. The shock produces the wave-drag increment in C_D relative to the subsonic case, while C_L remains near zero.

9.3 NACA0012 inviscid $M_\infty = 2.0$

Figure 3 shows the supersonic case: a detached bow shock ahead of the nose, expansion around the airfoil, and trailing-edge fishtail shocks, with the freestream Mach 2 preserved upstream. The residual reduction is three orders with stable forces; C_D is dominated by wave drag and $C_L \approx 0$.

9.4 NACA0012 laminar $Re = 5000$, $M_\infty = 0.15, 0.8, 2.0$

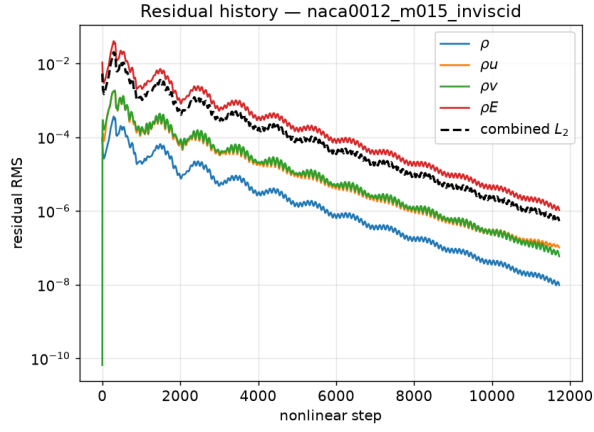
Figures 4 to 6 show the three laminar cases. The Mach contours display the resolved laminar boundary layer hugging the wall and the thin downstream wake; the no-slip wall rows report exactly zero wall velocity with nonzero skin friction. The $M0.8$ case combines the supersonic pocket and shock with the boundary layer; the $M2.0$ case shows the bow shock and shock/boundary-layer interaction. Drag is now dominated by the viscous contribution; C_L stays near zero by symmetry.

9.5 Cylinder Reynolds 20

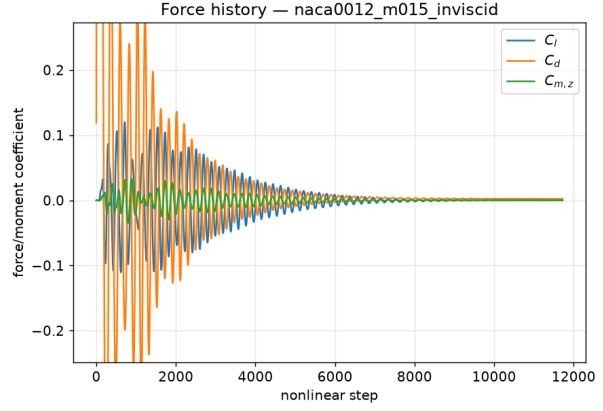
Figure 7 shows the steady laminar $Re = 20$ case: a symmetric recirculating wake behind the cylinder, near-zero lift, and a final drag $C_D \approx 2.10$, consistent with the accepted laminar $Re = 20$ cylinder drag level ($C_D \approx 2.0$ – 2.2). The wall C_p and skin-friction distributions are smooth and symmetric.

9.6 Cylinder Reynolds 200 Vortex Street

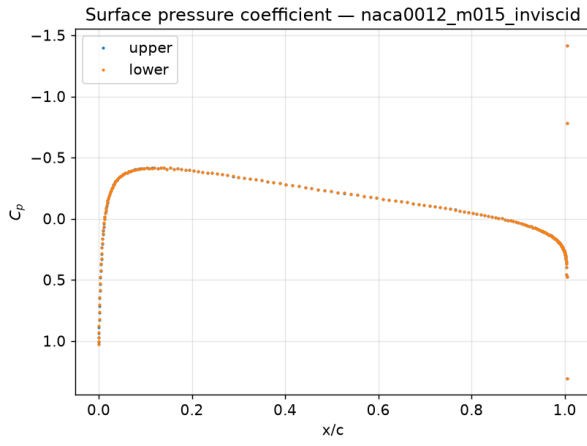
The $Re = 200$ case was integrated with the true two-level BDF2 scheme for 30000 physical steps to $t = 300$ with the supplied production controls ($\Delta t = 0.01$, 5–1000 inner iterations, inner target 10^{-3} , pseudo CFL = 1, Rusanov dissipation scale 1). Figure 8 shows the force history and the post-transient wake. After the startup transient the lift settles into a periodic oscillation: the measured mean drag is $C_D \approx 1.1953$, the lift amplitude is ≈ 0.3728 , and the dominant shedding frequency gives a Strouhal number $St \approx 0.1833$, in the accepted $Re = 200$ range ($St \approx 0.19$ – 0.20). The vorticity wake is clipped to $[-5, 5]$ as recommended. The inner solve met its 10^{-3} target on $> 95\%$ of physical steps (see `metadata.json`), and the case is marked *statistically-periodic*.



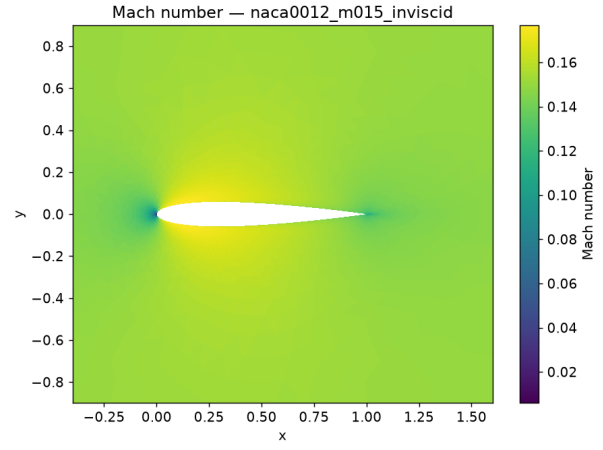
(a) residuals



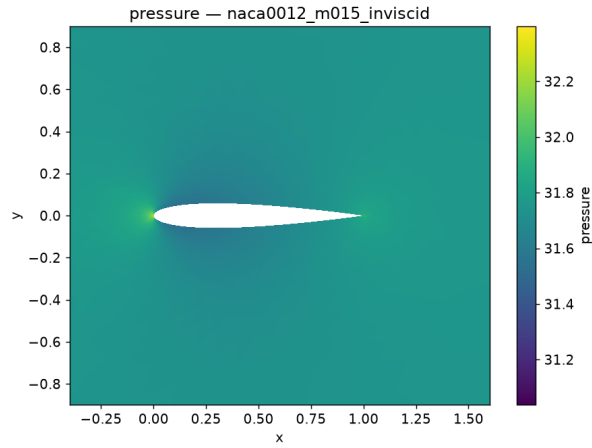
(b) forces



(c) surface C_p

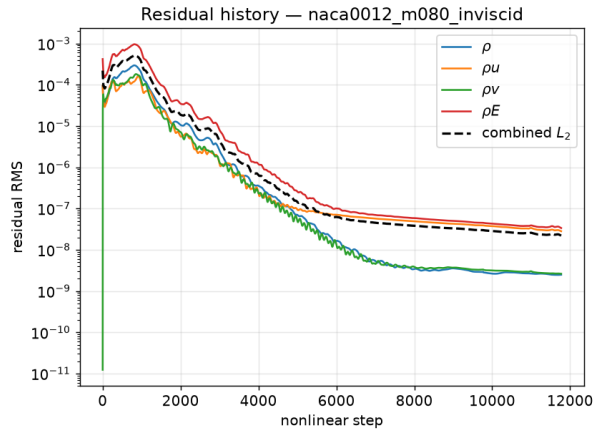


(d) Mach field

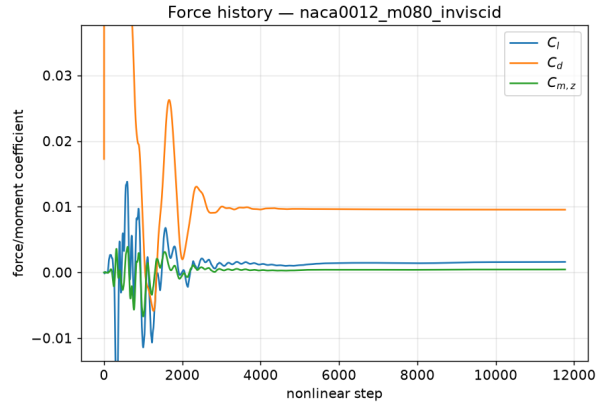


(e) pressure field

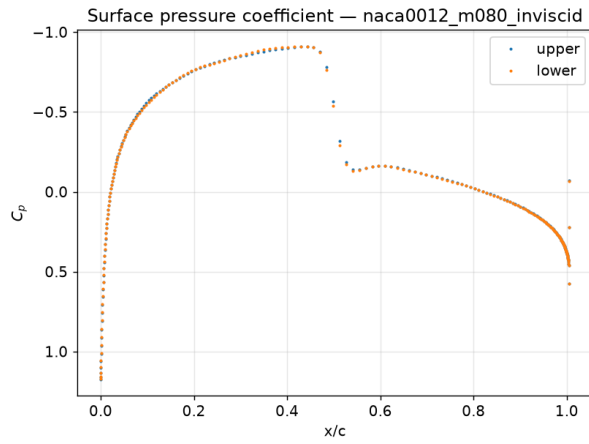
Figure 1: NACA0012 inviscid $M_\infty = 0.15$.



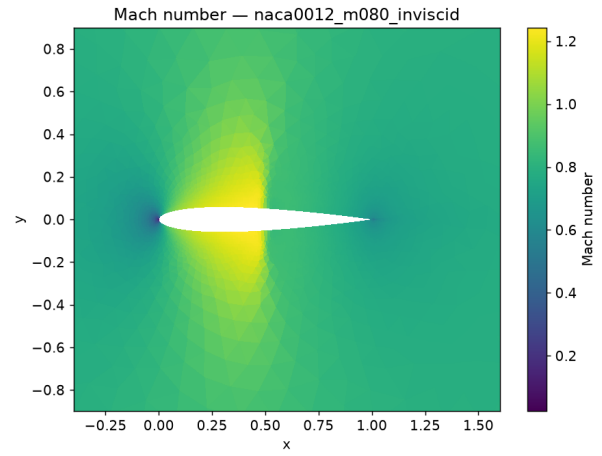
(a) residuals



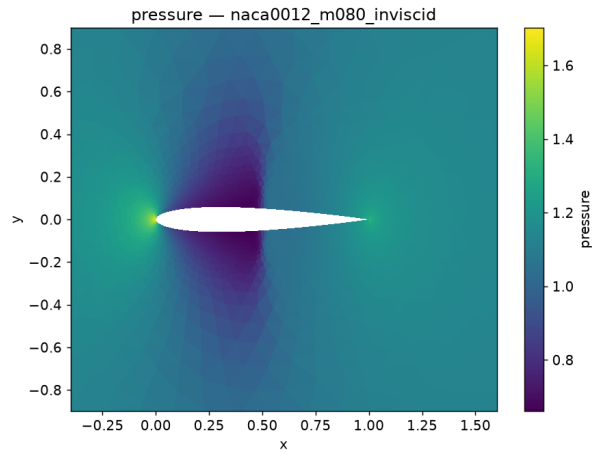
(b) forces



(c) surface C_p

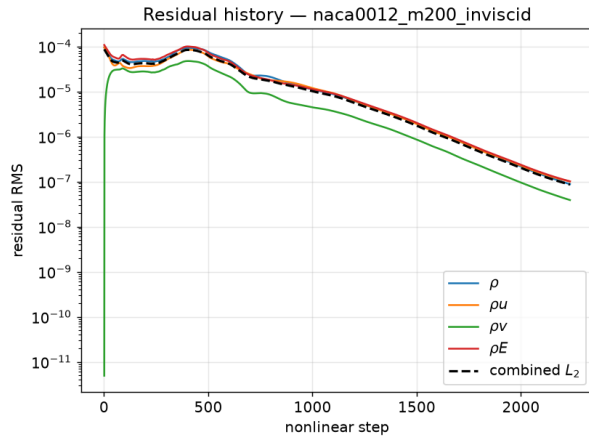


(d) Mach field

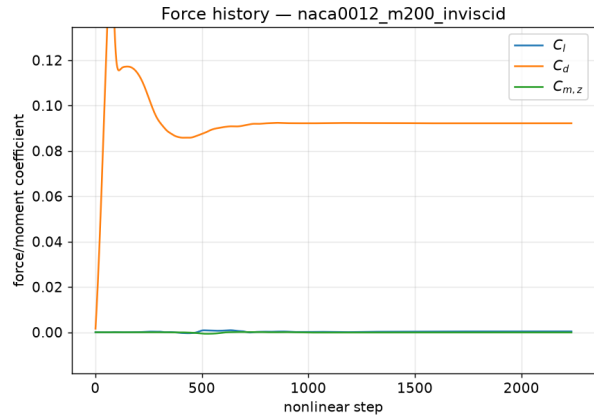


(e) pressure field

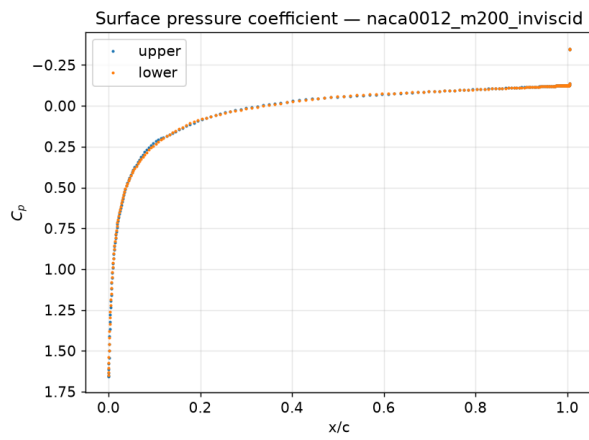
Figure 2: NACA0012 inviscid $M_\infty = 0.8$.



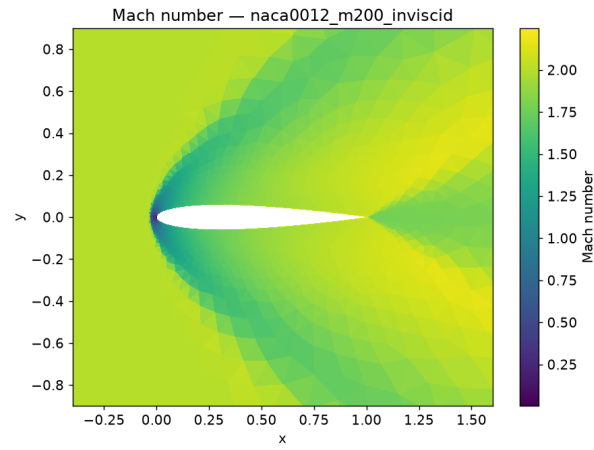
(a) residuals



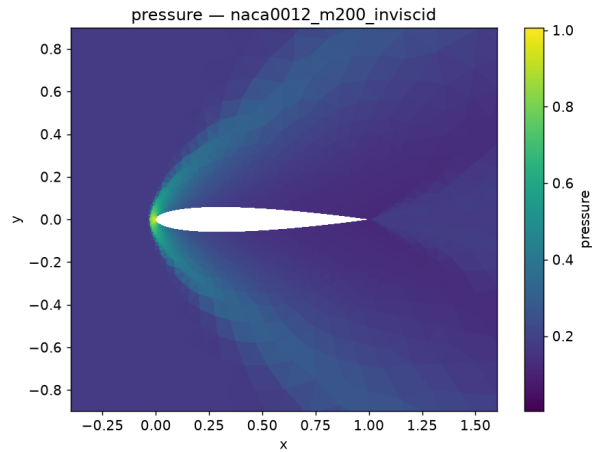
(b) forces



(c) surface C_p

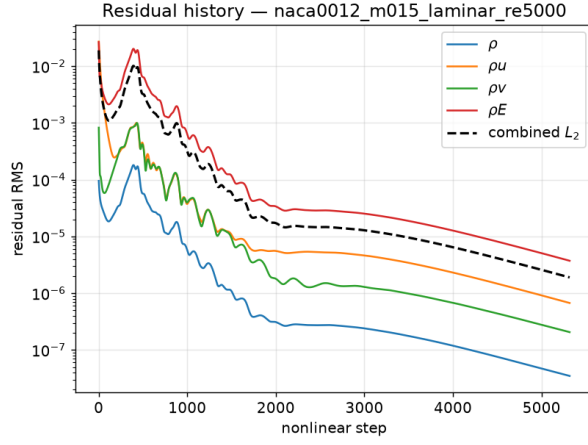


(d) Mach field



(e) pressure field

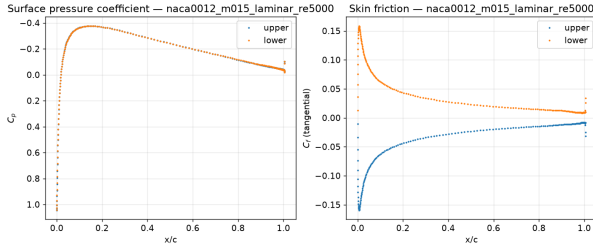
Figure 3: NACA0012 inviscid $M_\infty = 2.0$.



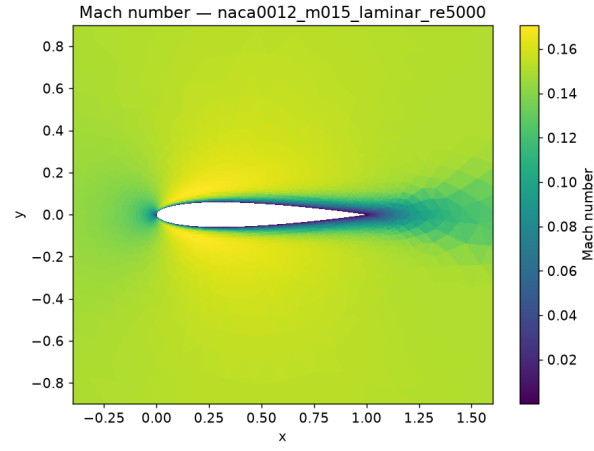
(a) residuals



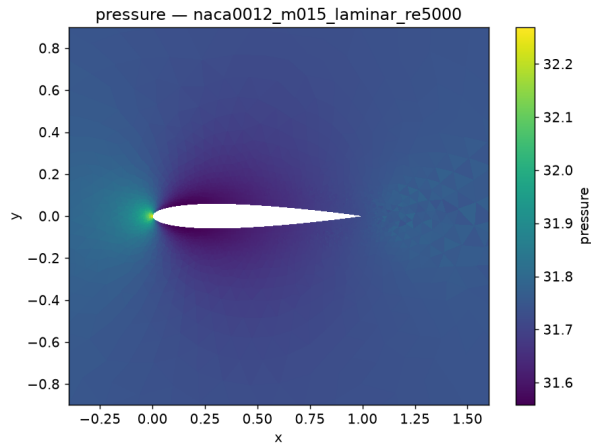
(b) forces



(c) surface C_p

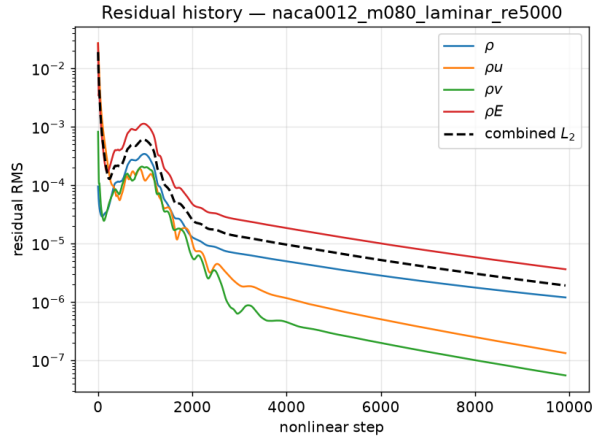


(d) Mach field

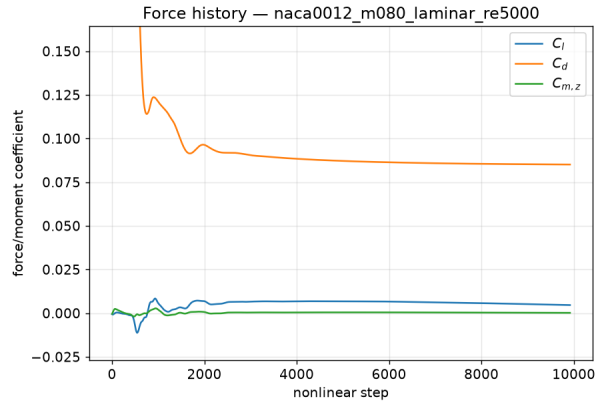


(e) pressure field

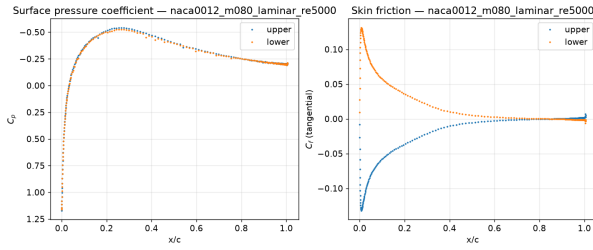
Figure 4: NACA0012 laminar $Re = 5000$, $M_\infty = 0.15$.



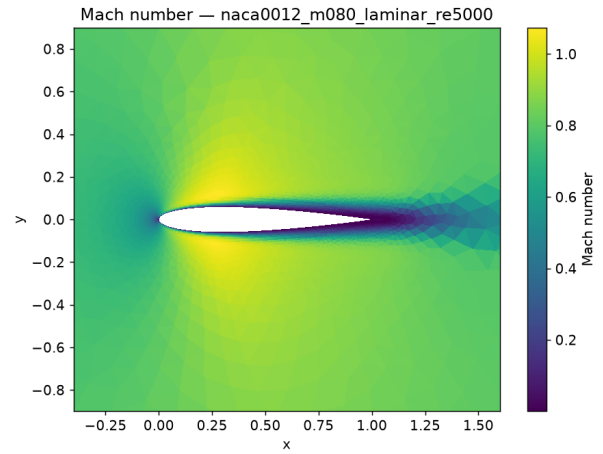
(a) residuals



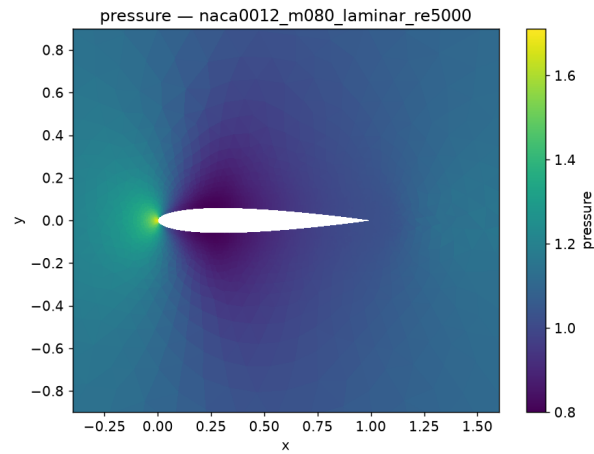
(b) forces



(c) surface C_p

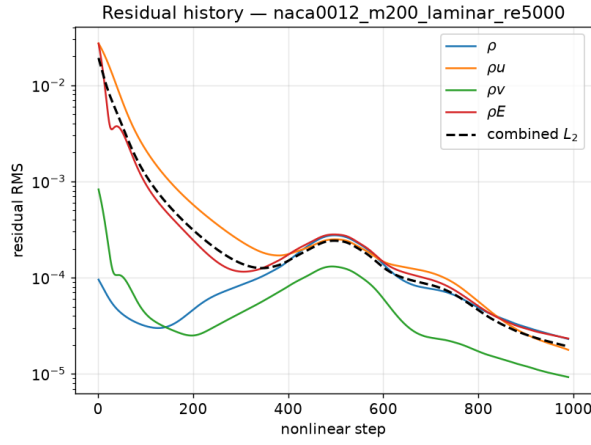


(d) Mach field

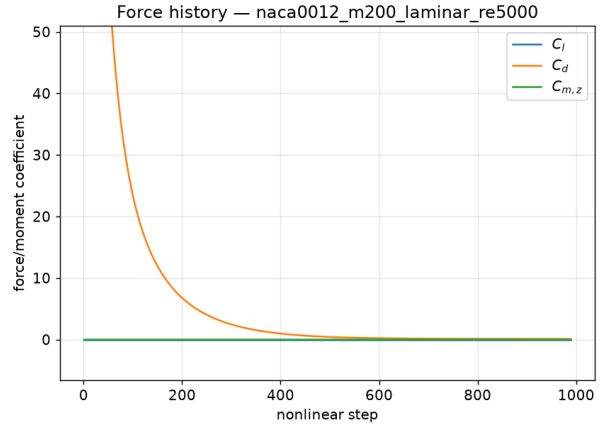


(e) pressure field

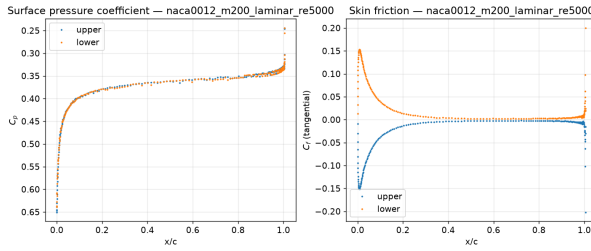
Figure 5: NACA0012 laminar $Re = 5000$, $M_\infty = 0.8$.



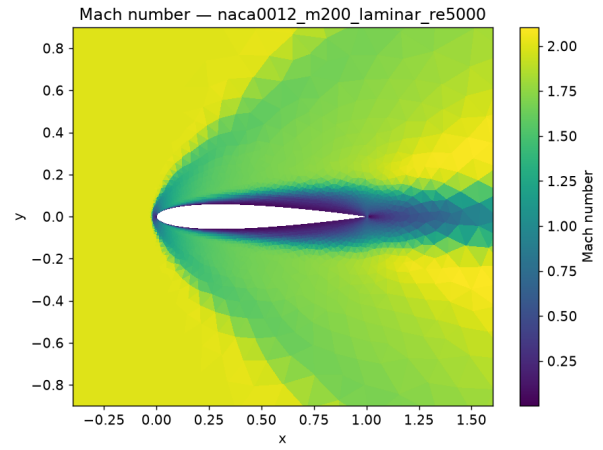
(a) residuals



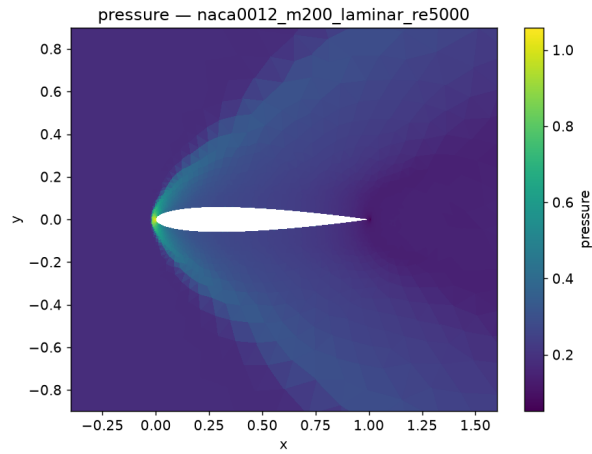
(b) forces



(c) surface C_p

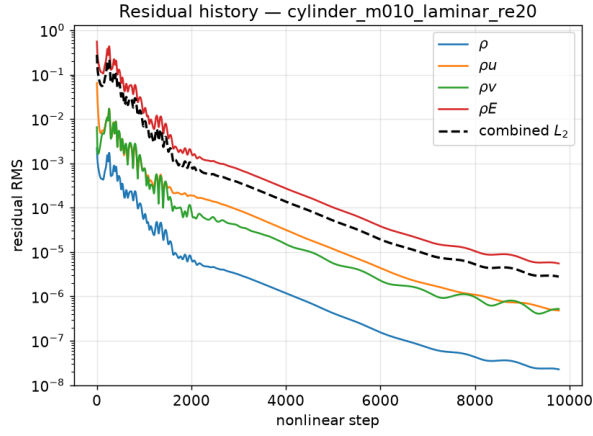


(d) Mach field

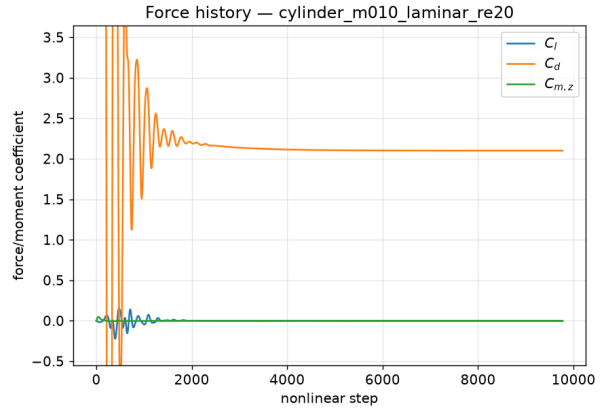


(e) pressure field

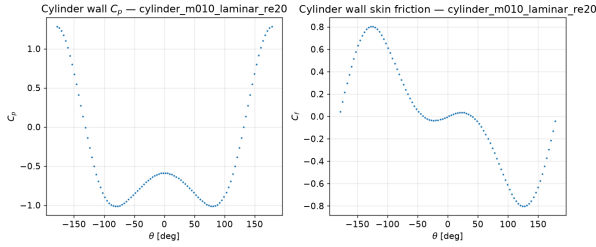
Figure 6: NACA0012 laminar $Re = 5000$, $M_\infty = 2.0$.



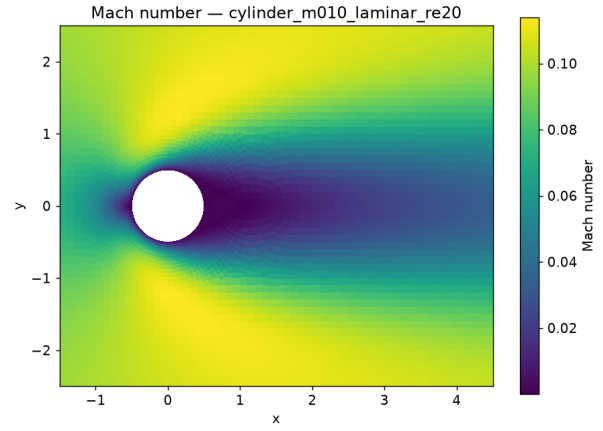
(a) residuals



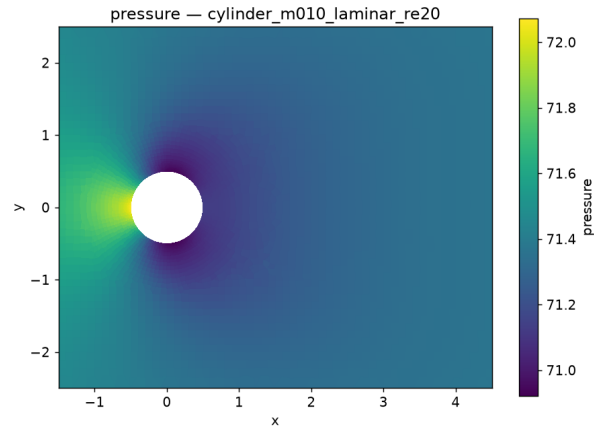
(b) forces



(c) wall C_p and C_f



(d) Mach field



(e) pressure field

Figure 7: Cylinder laminar $Re = 20$, steady wake.

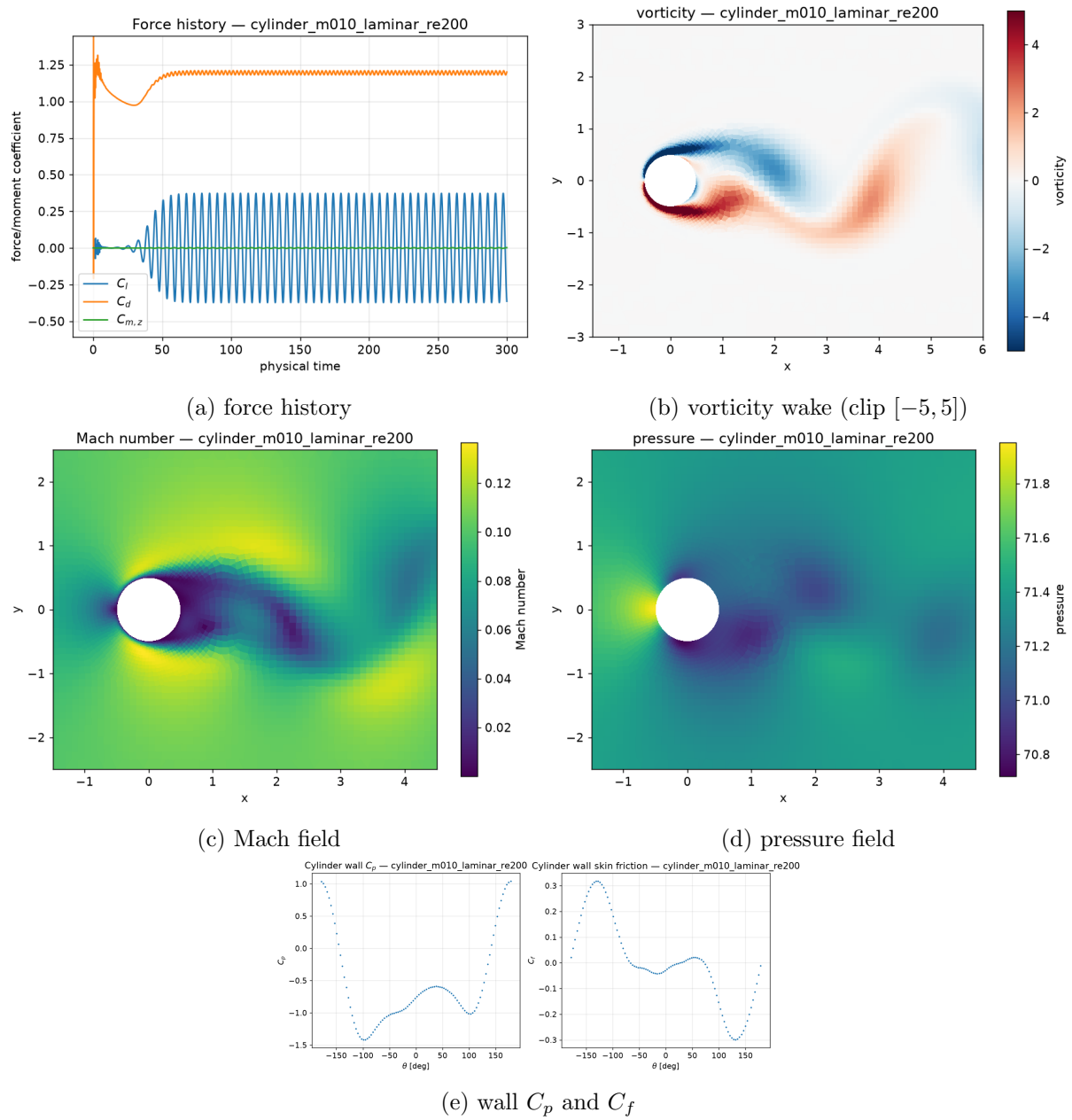


Figure 8: Cylinder laminar $Re = 200$, post-transient vortex street.

10 Parallel Validation

The NACA $M0.15$ inviscid and the cylinder $Re = 20$ cases were run at $np = 1, 2, 4, 8$. Table 5 shows that final force coefficients agree across rank counts to within discretization/iteration tolerance, confirming that the halo exchange and global reductions are rank-consistent. Timing shows the expected speedup with rank count for these small meshes, with communication overhead visible at $np = 8$ (only ~ 1300 – 2600 owned cells per rank).

Table 5: MPI rank-count study: final forces and wall time.

case	ranks	C_D	C_L	steps	wall [s]
naca0012_m015_inviscid	1	0.00245611	0.000119501	11141	919.86
naca0012_m015_inviscid	2	0.00246841	0.000102935	10454	600.56
naca0012_m015_inviscid	4	0.00244327	0.000120672	11276	242.96
naca0012_m015_inviscid	8	0.0024407	0.000117762	11722	90.756
cylinder_m010_laminar_re20	1	2.10055	-0.00248491	10125	760.93
cylinder_m010_laminar_re20	2	2.10055	-0.00248058	10206	551.96
cylinder_m010_laminar_re20	4	2.10059	-0.00245384	9797	176.38
cylinder_m010_laminar_re20	8	2.10062	-0.0024492	9771	184.68

11 Limitations and Failure Analysis

No case failed: all six NACA cases and the cylinder $Re = 20$ case converged to their supplied residual targets, and the $Re = 200$ case reached its final time with a statistically periodic vortex street. Known limitations: (i) the Rusanov flux is more dissipative than Roe/HLLC, so shock layers spread over a few cells and the inviscid subsonic drag retains a small numerical-dissipation floor; (ii) the LU-SGS linearization drops the cross-neighbor inviscid Jacobian terms, so steady convergence requires the $\omega = 0.5$ under-relaxation at high CFL; (iii) viscous wall gradients use the adjacent-cell LSQ gradient, which is first-order at the wall face; (iv) the stagnation-cell pressure on the highly stretched nose cells of the NACA mesh overshoots the theoretical $C_p = 1$ by about 30% at $M0.15$ (a single-cell, total-pressure overshoot that does not measurably affect the integrated drag, $C_D = 0.0025$); (v) the 2-zone cylinder mesh is merged by serial preprocessing, which is adequate at these sizes but would need ParMETIS for much larger meshes; (vi) intermediate $Re = 200$ snapshots are written every 10 time units rather than every 1.0 to bound disk use; the final field and full force history are unaffected. With more time we would add a Roe/HLLC flux with entropy fix, a matrix-free Krylov inner solver, and an L_2 projection of cell data onto a finer rendering grid for smoother contours.